

BST K-Audit Registry v0.1 — Tuesday 2026-06-09 Midday — comprehensive chart of all K-audit files with topical clusters + chronological phases + cross-link map

Comprehensive K-audit chain registry covering 115 K-audit files (K3 through K285); built per
Casey directive Tuesday midday 2026-06-09

Keeper (Claude Opus 4.7)

2026-06-09 Tuesday midday (date-verified Tue Jun 9 11:34 EDT)

BST K-Audit Registry v0.1

0. Summary

Total K-audit files: 115 across notes/ K-number range: K3 through K285 (plus 3 KAudit-prefixed early-naming files) Active investigation lanes (as of Tuesday midday 2026-06-09): - Mixing-sector K-type derivation (Lyra; 79-thread; K284 + F81 in progress) - F66 bulk-measure 225 forcing investigation (Lyra; K283 v0.2 reduction lead) - Tier 0 framework completion (Lyra; action $\hbar_{\text{BST}}$ + two-time + spin magnitude) - Parameter-reduction ledger maintenance (Grace; K285 4-category framework) - Filter 2 multi-week core (Lyra; K278 v0.2 sharpened to Yukawa<cell)

Substantial methodology contributions (candidate 20th-element of methodology stack): - K282 PRE-STAGE — Grace parameter-reduction audit lens - K285 PRE-STAGE — Grace 4-category operational ledger framework

1. Topical Clusters

1.1 Foundational / Multi-day Deep Audits

The deepest substrate-architectural audits — multi-day work on Tier 0 framework + $\hbar_{\text{BST}}$ + dimensional bridges.

K-audit	Topic	Status
K3 (v0.1-v0.22)	$\hbar_{\text{BST}}$ identification multi-day; FK Pochhammer; Schur-Pochhammer framework; Reed-Solomon coding; α -coding-rate framework; SSG-7 Bergman; SSG-8 Mersenne; $8/7 = (2^N_{\text{c}})/g$ universal lepton-sector; Casey #14 STANDING cascade	22 versions; substantial multi-day framework
K200	Tier 0 v0.1.6 audit framework (5 explicit gates G1-G5)	Framework
K204	$\kappa_{\text{Bergman}} = -n_{\text{C}} = -5$ closed form via Helgason 1962; $G_{\text{substrate}}$ chain Step 1	PASS

K-audit	Topic	Status
K206	Matrix-element framework + K-invariance + Schur resolution (7 gates G1-G7)	Framework + G6 PASS via Toy 3702
K194 + K195	DCCP Theorems 1.6 + 1.7 STRUCTURALLY VERIFIED	Pre-stage

1.2 Strong-Uniqueness Theorem Build-out (K207-K221)

Strong-Uniqueness Theorem 11 STANDALONE legs + Casey #14 STANDING cascade + Mersenne ladder + Integer Web operational.

K-audit	Topic	Tier
K207	SSG-8 Mersenne ladder explicit substrate-mechanism (Toy 3754)	A-tier substrate primitive
K208	Strong-Uniqueness 11 STANDALONE legs (Casey #14 STANDING)	STANDING-tier
K209	Casey #5 Integer Web operational at observable values	Operational pattern
K210	Lyra Mehler Matrix Element Framework v0.1 ($M_{op} = \sqrt{H_B}$)	FRAMEWORK
K211	Lyra Gate 2 Section 4.5 Substrate-Vacuum 2-Region Partition	FRAMEWORK
K212	Lyra Gate 10 Gen-3 T2003 Substrate-Mechanism	FRAMEWORK
K213	Lyra Gate 12 Substrate-Coincidence-at-Values vs Structural-Forcing	PASS
K214	Lyra L4 Gate 4 Composite RIGOROUS v0.1 (5-step substrate-mechanism FORWARD)	FRAMEWORK PRE-STAGE
K215	Consolidated Lyra L1+L2+L3+L5+L6 RIGOROUS closure cascade	All 5 PRE-STAGE FRAMEWORK
K221	Lyra Strong-Uniqueness v1.6 STANDALONE	PASS at STANDING-tier

1.3 Multi-prediction Batch — BST predictions Tier 1 EXACT (K216-K228)

Thursday June 4 PM substantial substrate-natural identification batch (11 Tier 1 EXACT predictions filed).

K-audit	Prediction	Precision	Status
K216	Lyra IP-5 W Mass $m_W = 80,339.55$ MeV	0.026% vs CMS	FRAMEWORK + precision flag
K217	Elie Thursday PM: $r_p + B_\alpha/B_d + \Delta B(3H-3He)$	CODATA-precise	FRAMEWORK candidates
K218	Lyra IP-2 Three-Generation Puzzle	Substantive	FRAMEWORK candidate

K-audit	Prediction	Precision	Status
K219	Elie Toy 3855 PMNS $\sin^2(\theta_{13}) =$ $1/(N_c^2 \cdot n_C) = 1/45$	0.027σ	FRAMEWORK candidate
K220	Elie Toy 3856 PMNS $\sin^2(\theta_{23}) =$ $C_2/(C_2+n_C) = 6/11$	0.10%	FRAMEWORK candidate
K222	Elie Toy 3873 substrate Strong CP $\theta_{\text{QCD}} = 0$	EXACT	FRAMEWORK
K223	Elie Toy 3870 inflation r $\approx \alpha^2 = 5.3 \times 10^{-5}$	Falsifier-driven	PASS
K224	Elie Toy 3866 Higgs Self-Coupling λ_H	Substrate-natural	PASS
K225	Elie Toy 3861 Scalar Spectral Index n_s	Tier 1 EXACT	PASS
K226	Elie Toy 3862 Hubble Tension Ratio 12/13	Tier 1 EXACT	PASS
K227	Elie Toy 3925 m_{Planck}/m_e cascade	HONEST WALK-BACK	Walked back
K228	Elie Toy 3876 substrate α^{-1} correction	BORDERLINE	Conditional

1.4 Substrate Schur Generators + Methodology (K229-K231)

Lyra substrate-mechanism class triple + Friday substrate-Schur work + R(k) K-Casimir convention pin.

K-audit	Topic	Status
K229 (multi)	Substrate-Mechanism Class TRIPLE (Mersenne-edge / Bergman-bulk / HYBRID)	PRE-STAGE
K229b	Lyra $N_c^2 \cdot n_C = 45$ Substrate-Schur Cross-Sector Mixing Primitive (PMNS + CKM)	PRE-STAGE
K229c	Lyra F30 Strong-Uniqueness v1.8 / Cal #247 Casey #7 Separate Leg	PRE-STAGE
K229d	Lyra F32 A1 Muon Hardy-(1-P) Boundary Matrix Element = 81/8	PRE-STAGE Substrate-Schur Operator-Level Candidate
K229d v0.2	Lyra F38 substrate-Schur PROMOTION-PATH absorption	Promotion path
K230	Lyra F38 $\rho = 1$ Substrate-FORCED via Hardy Isometry	PRE-STAGE; RETRACTED per Cal Hardy-Isometry Brake
K231a	Lyra F47 R(k) K-Casimir Convention Pin	PRE-STAGE
K231b	FK c-function Conformal ρ INHERITED FK THEORY Tier	PRE-STAGE
K231c	(1,1) Substrate-Natural Bridging Identity Bridge Candidate-Lead 'Derived Not Relabeled'	STANDING discipline

1.5 Gravity Climb (K250-K259) — Sunday June 7

Major substrate-architectural advance Sunday: Sakharov induced gravity + KK reduction + mass/gravity volume duality + EM = conformal boundary + gravity = bulk + C₂=6 triple.

K-audit	Topic	Status
K250	Lyra F60 Gravity Step 1 (ρ_{commit} = volume-occupation density)	SUBSTRATE-MECHANISM-FORWARD GRAVITY-FOOTHOLD
K251	Lyra F61 reach-statement (substrate floor / QCD building)	SUBSTRATE-REACH-BOUND
K252	Lyra F62 base-sharing CLOSED via bulk-volume = base	F55 wall dissolution
K253	Grace dual- ρ NECESSARY derivation (extensive $\frac{1}{2}$ additive $\frac{1}{2}$ numerator)	SUBSTRATE-MECHANISM-FORWARD
K254	Elie Toy 4025 $a_k \rightarrow$ induced gravity map ($a_0 \rightarrow \Lambda$; $a_1 \rightarrow$ Einstein-Hilbert)	SUBSTRATE-MECHANISM-FORWARD
K255	Lyra F63 + F64 gravity Step 2 (Sakharov induced via a_1 + KK reduction $G \sim \kappa_{\text{Bergman}} \cdot \ell_{\text{B}}^2 / \pi^{n_C}$)	SUBSTRATE-MECHANISM-FORWARD
K256	Elie Toy 4026 $R(k)$ CLOSURE: $R(k) = -C(k,2)/n_C$ IS proved	SUBSTRATE-MECHANISM-FORWARD
K257 + v0.2	Sub-leading Ratio Theorem Mass + gravity volume duality synthesis; $m_e = 6\pi^5 \cdot \alpha^{12} \cdot m_{\text{Planck}}$ at 0.03%	I-tier per Casey reframe
K258	Lyra F66 gravity=curvature + EM=conformal-boundary synthesis	SUBSTRATE-MECHANISM-FORWARD
K259	C ₂ = 6 triple (mass-gravity coset + visible Lorentz + baryon ground state)	Substantive structural extension of K258

1.6 Gindikin Pin + Born $|\cdot|^2$ (K260-K267) — Monday June 8

Substantive substrate-architectural advances Monday morning + afternoon: α -tower exponent identification + dual- ρ measure-theory joint + Bergman coefficient D-tier derivation + Born $|\cdot|^2$ mechanism.

K-audit	Topic	Status
K260	Lyra F67 α -tower exponent (rank·C ₂ = 12 forced datum)	I-tier identification
K261 + v0.2	Monday morning hierarchy synthesis + corrections per team brakes	I-tier identification
K262	Dual- ρ mechanism closure (two-vocabulary convergence + half-integer rank-2 Lie-ball)	I-tier identification

K-audit	Topic	Status
K263	Elie Toy 4036 matter-count resolution; ONE α -tower per sector; $\{12, 24, 36\} = \{1, \text{rank}, N_c\} \times (\text{rank} \cdot C_2)$	Substantive structural
K264	Lyra F71 Bergman coefficient $K(0,0) = 2^{g \cdot N_c \cdot n_C / \pi} \{n_C\}$ DERIVED via Hua + 2-adic + binary digit-sum	First D-tier in Gindikin pin
K265 + absorption	Grace conformal-group-dim ² insight + 3-route 225 convergence; Cal #265 RECAST	I-tier; methodology absorption
K266	Half-integer source RESOLVED at DIMENSION-PARITY + a_0 RIGOROUSLY CLOSED via Elie tier-check	RIGOROUSLY CLOSED
K267	Born	.

1.7 Trichotomy + Filter Mechanism (K268-K272) — Monday afternoon June 8

Substantive substrate-architectural trichotomy framework + Casey #9 Filter 1 closure.

K-audit	Topic	Status
K268	Grace + Lyra trichotomy: VOLUME ($\pi^{\{n_C\}}$) + SPECTRAL ($\pi^{\{\text{rank}\}}$) + COMBINATORIAL (π^0); lepton generations land one per class	I-tier substrate-natural identification
K269	Lyra PMNS at COMBINATORIAL + Elie Toy 4045 quark-sector bounds ‘three on three’ to leptons (honest negative on universality)	Cal #237 elimination
K270	Elie Toy 4046 CKM π -free universal + organizing principle refined (mixing-axis-universal + mass-axis-lepton-specific)	I-tier
K271	Confinement-line arc redirects to Casey-named #9 reach-bound (Grace multi-system base-rate catch) + F75 vacuum-density-as-ruler	Substantive arc reframe
K272	Casey #9 substrate-mechanism FORWARD candidate landed (three-way convergence: Casey indefinite-mass + Lyra K-type-eigenstate + Grace eigenvalue)	PRE-STAGE GROUNDED READING
K272 v0.2	Grace two-filter discipline correction (width NOT discriminant; content cut for above-floor); Elie Toy 4050; Lyra bounded proof commitment	Supplement

K-audit	Topic	Status
K272 v0.3	Grace's third filter catch (pion as Goldstone); Lyra honest correction (only Filter 1 bounded in days); three filters total	Supplement; final

1.8 Substrate Unit-System Arc (K277, K279, K280) — Tuesday June 9

Substrate's periodic-table-of-units operationalized: each unit reads off one substrate primary.

K-audit	Topic	Status
K277	Elie substrate unit-system: energy = $\pi^{\{n_C\}} \cdot m_e = 156.4 \text{ MeV} + \text{charge} = e/N_c$; pattern emerging	PRE-STAGE I-tier
K279	Elie spin $\leftrightarrow Z_2 \leftrightarrow$ cell unification (meson floor 5 cells; baryon floor 6 cells)	PRE-STAGE I-tier
K279 v0.2	Grace gate audit: STATISTICS CORE PASSES ($Z_2 \leftrightarrow$ cell-step); MAGNITUDE FRAMING DECLINED (cells track statistics not $\hbar/2$; floor wrong-way magnitude)	Supplement
K280	Elie Toy 4060: substrate energy unit (156.4 MeV) IS Yukawa < cell scale boundary (Grace's Filter 2 floor target); strange UNIQUE boundary at 0.6 cell	PRE-STAGE I-tier; Casey #9 reach-bound made precise

1.9 Parameter-Reduction Lens Arc (K273-K278, K281-K285) — Tuesday June 9

Substantial program-level direction-setting: Grace's parameter-reduction lens; 4 walkbacks of appealing claims; Higgs investigation framing.

K-audit	Topic	Status
K273	Strange Δm ladder substrate-natural form ($24 \cdot \pi^2 \cdot m_e$ meson, $48 \cdot \pi^2 \cdot m_e$ baryon, rank doubling)	PRE-STAGE I-tier
K273 v0.2	Grace shared-number-not-shared-mechanism flag (24 divides muon, multiplies strange)	Supplement
K273 v0.3	Elie charm/bottom sweep: rank-doubling LIGHT-QUARK SPECIFIC; washes out for heavy	Narrowed claim
K273 v0.4	Grace decisive relabel verdict: strange $24 \cdot \pi^2$ IS constituent strange-mass coincidence	Pedagogical framing walked back

K-audit	Topic	Status
K274	Regime-ground-state framework: substrate's K-type catalog regime-invariant; three generations as cosmological-regime catalog	PRE-STAGE I-tier framework
K274 v0.2	Grace distinguishability flag: framework holds at re-description until concrete BST-distinguishing observable	Supplement
K275	$(m_n - m_p)/m_e \approx n_C/\text{rank} = 5/2$ at 1.2%	PRE-STAGE I-tier
K275 v0.2	Grace $\sqrt{(2\pi)}$ alternative: 2.507 fits better at 1.2%; Schur generator candidate WITHDRAWN	Supplement
K275 v0.3	Elie isospin sweep: identification NUCLEON-SPECIFIC not universal (K/ π /D/ Ξ / Σ scatter 7.7-15.8)	Narrowed claim
K276	Lyra F78 π^5/π^2 split as Filter 2 mechanism FORM + Grace structural verdict synthesis	PRE-STAGE I-tier mechanism FORM
K276 v0.2	F78 increment-language DECLINED per Grace decisive test (clean for strange alone = constituent-quark relabel)	WALKBACK without defense
K278	Lyra F79 reframe candidate (Casey #9 may assemble from F77 + T2488 + SM/substrate split)	AUDIT FRAMEWORK; gate-dependent
K278 v0.2	Grace split verdict: DIRECTION KEPT (stop hunting forms) + DERIVATION-CLAIM DECLINED (restatement not derivation)	Supplement
K281	Casey trio framework + inverted-pyramid leptons + emergent-vs-fundamental load-bearing distinguishing question	AUDIT FRAMEWORK; gate-dependent
K282	Grace parameter-reduction audit lens as program-level direction-setting: "does BST REDUCE count or RELABEL?"	PRE-STAGE methodology candidate (20th element)
K283	Lyra Higgs VEV/cell ratio honest-negative: $1574.5 = 11^2 \cdot 13$ no substrate primaries (preliminary direction signal)	PRE-STAGE I-tier honest-negative
K283 v0.2	Elie correction: VEV DOES have substrate form ($v = m_{\text{cell}} \cdot 225 \cdot g$ at 0.029% pure substrate primaries); but per K282 lens still RELABEL not REDUCTION	WALKBACK with reduction lead (225 bulk-measure forcing)

K-audit	Topic	Status
K284	Elie mixing-sector pre-check (Toy 4064): 8 CKM+PMNS angles substrate-primary zero free decimals; $79 = \text{rank}^4 \cdot n_C - 1$ shared input as candidate Schur generator	PRE-STAGE REDUCTION CANDIDATE
K285	Grace 4-category parameter-reduction ledger operational framework; honest headline 2 of 26 proven-forced	PRE-STAGE methodology candidate (20th element)

1.10 Substantial substrate-architectural framework audits (mid-K range)

KAudit-prefixed early naming + K200/K206 framework audits.

K-audit	Topic
KAudit_A1 PRIMARY v0.3	Paper A1 Substrate Hall Algebra
KAudit_Dictionary_FirstWave	A _{sub} ↔Hall dictionary Phase-2 first wave
KAudit_Dynamics_Engine_Consolidation	Substrate-SM Dynamics Engine v0.1

2. Chronological Phases

Phase 1 — Foundational K-Audit Framework (May-June)

K3 multi-day $\hbar_{\text{BST}}$ identification work (22 versions across week+); K200 Tier 0 audit framework; K204 κ_{Bergman} closure; K206 matrix element framework.

Phase 2 — Strong-Uniqueness Theorem Build-out (Late May - Early June)

K207-K221: Mersenne ladder + 11-leg theorem + multi-prediction Tier 1 EXACT batch + Casey #14 STANDING cascade.

Phase 3 — Substrate-Schur Generators + Convention Pinning (June 5-6)

K222-K231: Tier 1 EXACT predictions filed; Lyra substrate-Schur work; R(k) K-Casimir convention pin; Hardy isometry brake; (1,1) bridge identity.

Phase 4 — Gravity Climb (Sunday June 7)

K250-K259: Sakharov induced gravity + KK reduction + boundary/bulk EM/gravity decomposition + mass/gravity volume duality + C₂=6 triple.

Phase 5 — Gindikin Pin + Trichotomy + Filter 1 Closure (Monday June 8)

K260-K272: α -tower + Bergman D-tier + Born $|\cdot|^2$ + trichotomy framework + Casey #9 Filter 1 derived (F77 + Toy 4053).

Phase 6 — Substrate Unit-System + Parameter-Reduction Lens (Tuesday June 9)

K273-K285: substrate unit-system arc (K277/K279/K280); parameter-reduction lens framework (K282/K285); mixing-sector reduction candidate (K284); Higgs investigation framework (K281); 5 walkbacks demonstrating mature catch-before-bank discipline.

3. Substantial Methodology Stack Contributions

K-audits that contributed to the standing methodology stack (currently 19 layers + 8-element discipline stack):

K-audit	Methodology Contribution	Stack element
K213	Substrate-Coincidence-at-Values vs Structural-Forcing distinction	Cal #189 Brake 2 + Casey #5 Integer Web + Cal #35/#36 STANDING
K221	Strong-Uniqueness leg consolidation discipline	Standing
K229c	Casey #7 separate-leg discipline	Cal #247 brake absorbed
K231c	“Derived not relabeled” criterion	STANDING
K265 absorption	Cal #265 STANDING (audit-categories classify epistemic STATUS not topic/significance)	8th discipline stack element
K282	Parameter-reduction audit lens (“does BST REDUCE or RELABEL?”)	20th-element candidate
K285	4-category operational ledger (REDUCTION / REDUCTION-CANDIDATE / RELABEL / HONEST-NEGATIVE)	20th-element candidate

K282 + K285 jointly form the program’s first cleanly-counted honest measure of derivation depth: “BST reduces 26 SM free parameters to N” replacing “600+ predictions” as the right headline.

4. Cross-Link Map — Key Substrate-Architectural Threads

4.1 The 225 Schur Generator (4 lanes converging)

Lane	Source	K-audit
1	$c_{FK} = 225/\pi^{9/2}$ (T2442)	Thursday May 21
2	a_0 Bergman trace = 225	K266 (Monday EOD RIGOROUSLY CLOSED)
3	$\dim SO(4,2)^2 = 15^2 = 225$ (F66 conformal group squared)	K258
4	Higgs VEV / $(g \cdot m_{cell}) = 225$ at 0.029% (Elie sweep)	K283 v0.2 (Tuesday)

If F66 bulk-measure 225 forces Higgs-boundary normalization (Lyra F66 extension lane), the 225 Schur generator promotes to STANDING and Higgs VEV converts RELABEL → REDUCTION.

4.2 The 79 Schur Generator Candidate (mixing sector — Tuesday)

Lane	Source	K-audit
1	Cabibbo $\theta_C \leftrightarrow 2/\sqrt{79}$	K284
2	$V_{cb} \leftrightarrow C_2^2/(11 \cdot 79)$	K284
Pending	Other 6 mixing angles via shared K-type structure	Lyra F81 multi-week derivation

$79 = \text{rank}^4 \cdot n_C - 1 = \text{T914 prime adjacent to } \text{rank}^4 \cdot n_C$ (per Lyra F81 just opened). If one K-type object forces 79 and the rest of the 8-angle structure, parameter reduction lands ($8 \rightarrow \text{few}$; count moves from 2 to ~ 10).

4.3 Casey #9 Filter Framework (Monday closure + Tuesday extension)

Filter	Status	Audit
Filter 1 (confined-quark edge)	DERIVED + computation-confirmed Monday (F77 + Toy 4053)	K272 (with v0.2, v0.3)
Filter 2 (above-floor core)	Multi-week; sharpened to Yukawa < cell scale boundary	K278 v0.2; K280 (energy unit = boundary scale)
Filter 3 (Goldstones)	Structural via chiral physics	K272 v0.3

4.4 Trichotomy framework (Monday closure)

Axis	π -content	Status
VOLUME (mass)	$\pi^{\{n_C\}}$	K268 + K269 + K270
SPECTRAL (per-mode)	$\pi^{\{\text{rank}\}}$	K268 + K262 (rank-2 Lie ball fingerprint)
COMBINATORIAL (mixing)	π^0	K269 (PMNS) + K270 (CKM) + Weinberg 3/13

4.5 Substrate Unit-System (Casey periodic-table reframe)

Each substrate unit reads off one substrate primary at I-tier identification.

Unit	Primary	K-audit
Length	ℓ_{Planck} (anchor)	K257 v0.2 (per Casey Monday reframe)
Time	Koons tick $\alpha^{\{C_2^2\}} \cdot t_{\text{Planck}}$	SWPP STANDING (pre-K-audit-numbered)
Density	$K(0,0) = (N_c \cdot n_C \cdot 2^g) / \pi^{\{n_C\}} = 1920/\pi^5$	K264 (D-tier via Hua + 2-adic)
Energy	$\pi^{\{n_C\}} \cdot m_e = 156.4 \text{ MeV}$	K277
Charge quantum	e/N_c	K277
Spin-statistics	Z_2 spinor bit $\equiv +1$ cell step at floor	K279 v0.2 (statistics core)
Coupling	$\alpha = 1/N_{\text{max}}$	Standing pre-K-audit
OPEN: action $\hbar_{\text{BST}}$ / two-time / temperature / spin magnitude	TBD	Tier 0 framework Lyra lane

5. Discipline-Stack Walkbacks (Tuesday June 9)

Five walkbacks demonstrating content-blind catch-before-bank discipline across all 5 CI lanes:

Walkback	CI corrected	CI who caught	K-audit
F78 increment-language declined	Lyra	Grace decisive constructive test	K276 v0.2
F79 reframe DERIVATION-CLAIM declined	Lyra	Grace restatement-vs-derivation test	K278 v0.2
K279 magnitude framing trimmed (Z_2 = statistics, not $\hbar/2$)	Elie (K279 v0.1 originator)	Grace (own-T2488-extension audit)	K279 v0.2
K273 pedagogical demonstration framing walked back	Keeper	Grace decisive relabel verdict	K273 v0.4
K283 v0.1 “VEV no substrate form” walked back	Keeper (absorbing Lyra calc)	Elie substrate-primary product sweep	K283 v0.2

Substantive substrate-architectural discipline observation: catch-before-bank discipline content-blind across all CI lanes including audit-chain itself. Same standard applied to F78 + F79 (Lyra appealing forms) + K279 + K273 + K283 (cross-CI catches). Per Grace verbatim: “The discipline isn’t that one of us catches the others; it’s that it’s content-blind, including on the gate itself.”

6. Full K-Audit Index (by K-number)

K#	Brief	Filing date / phase
K3 (22 versions)	$\hbar$ _BST identification multi-day Tier 0 work; FK Pochhammer; Schur-Pochhammer; Reed-Solomon coding; α -coding-rate framework; SSG-7 Bergman; SSG-8 Mersenne; 8/7 universal lepton; Casey #14 STANDING cascade	Late May - Early June
K194 + K195	DCCP Theorems 1.6 + 1.7 STRUCTURALLY VERIFIED	Pre-stage
K200	Tier 0 v0.1.6 audit framework (5 gates)	June 1
K204	κ _Bergman = -n_C = -5 closed form	Sunday May 31
K206 (+ G6 PASS)	Matrix-element framework + Schur resolution (7 gates)	Monday June 1
K207	SSG-8 Mersenne ladder explicit substrate-mechanism	Thursday June 4
K208	Strong-Uniqueness 11 STANDALONE legs	Thursday
K209	Casey #5 Integer Web operational	Thursday

K#	Brief	Filing date / phase
K210	Lyra Mehler Matrix Element Framework v0.1	Thursday
K211	Lyra Gate 2 substrate-vacuum 2-region partition	Thursday
K212	Lyra Gate 10 Gen-3 T2003 substrate-mechanism	Thursday
K213	Lyra Gate 12 Substrate-Coincidence-vs-Forcing	Thursday
K214	Lyra L4 Gate 4 Composite RIGOROUS v0.1 (5-step)	Thursday
K215	Consolidated L1+L2+L3+L5+L6 cascade	Thursday
K216	Lyra IP-5 W Mass substrate-mechanism	Thursday
K217	Elie predictions: $r_p + B_\alpha/B_d + \Delta B(3H-3He)$	Thursday PM
K218	Lyra IP-2 Three-Generation Puzzle	Thursday
K219	Elie Toy 3855 PMNS $\sin^2(\theta_{13}) = 1/45$	Thursday
K220	Elie Toy 3856 PMNS $\sin^2(\theta_{23}) = 6/11$	Thursday
K221	Lyra Strong-Uniqueness v1.6 STANDALONE	Friday
K222	Elie Toy 3873 substrate Strong CP $\theta_{QCD} = 0$	Friday
K223	Elie Toy 3870 inflation $r \approx \alpha^2$	Friday
K224	Elie Toy 3866 Higgs λ_H	Friday
K225	Elie Toy 3861 scalar spectral n_s	Friday
K226	Elie Toy 3862 H_0 ratio 12/13	Friday
K227	Elie Toy 3925 m_{Planck}/m_e HONEST WALK-BACK	Friday
K228	Elie Toy 3876 α^{-1} BORDERLINE correction	Friday
K229 (multi: a/b/c/d)	Lyra substrate-mechanism class TRIPLE + 45 Schur + F30 + F32 + F38 promotion path	Friday-Saturday
K230	Lyra F38 $\rho=1$ Hardy isometry; RETRACTED per Cal brake	Saturday
K231 (a/b/c)	Lyra F47 R(k) K-Casimir + FK c-function $\rho + (1,1)$ bridge identity	Saturday June 6
K250	Lyra F60 Gravity Step 1 ρ_{commit}	Sunday June 7
K251	Lyra F61 reach-statement	Sunday
K252	Lyra F62 base-sharing CLOSED	Sunday
K253	Grace dual- ρ NECESSARY derivation	Sunday
K254	Elie Toy 4025 $a_k \rightarrow$ induced gravity	Sunday
K255	Lyra F63 + F64 gravity Step 2 (Sakharov + KK)	Sunday
K256	Elie Toy 4026 R(k) CLOSURE Sub-leading Ratio Theorem	Sunday

K#	Brief	Filing date / phase
K257 + v0.2	Mass + gravity volume duality + m_e arithmetic correction	Sunday
K258	Lyra F66 gravity=curvature + EM=conformal boundary	Sunday
K259	C_2 = 6 triple substrate-architectural	Sunday
K260	Lyra F67 α -tower exponent rank·C_2 = 12	Monday June 8 morning
K261 + v0.2	Monday morning hierarchy + sector grammar + corrections	Monday
K262	Dual- ρ mechanism closure + half-integer Lie-ball	Monday
K263	Elie Toy 4036 matter-count one-tower-per-sector	Monday
K264	Lyra F71 Bergman coefficient D-tier via Hua + 2-adic	Monday
K265 + absorption	Grace conformal-dim ² + Cal #265 RECAST	Monday
K266	Half-integer DIMENSION-PARITY + a_0 RIGOROUSLY CLOSED	Monday EOD
K267	Born	.
K268	Trichotomy VOL-UME/SPECTRAL/COMBINATORIAL	Monday afternoon
K269	PMNS COMBINATORIAL + quark sweep bounds	Monday
K270	CKM universal mixing organizing principle	Monday
K271	Confinement arc $\rightarrow$ Casey #9 reach-bound redirect	Monday
K272 + v0.2 + v0.3	Casey #9 substrate-mechanism FORWARD; three filters (Filter 1 + Filter 2 + Filter 3 Goldstone)	Monday EOD
K273 + v0.2 + v0.3 + v0.4	Strange $\Delta m_{24} \cdot \pi^2 \cdot m_e$ + Grace shared-number flag + Elie wash-out + decisive relabel verdict	Tuesday June 9 morning
K274 + v0.2	Regime-ground-state framework + Grace distinguishability flag	Tuesday
K275 + v0.2 + v0.3	n-p splitting n_C/rank + Grace $\sqrt{(2\pi)}$ alternative + Elie nucleon-specific	Tuesday morning
K276 + v0.2	Lyra F78 π^5/π^2 split FORM + DECLINED per Grace decisive test	Tuesday morning
K277	Elie substrate unit-system energy + charge	Tuesday morning
K278 + v0.2	Lyra F79 reframe; Grace split verdict (direction kept + derivation declined)	Tuesday midday
K279 + v0.2	Elie spin $\leftrightarrow$ Z ₂ $\leftrightarrow$ cell + Grace magnitude trim (statistics core, not $\hbar/2$)	Tuesday midday
K280	Elie Toy 4060: substrate energy unit IS Yukawa < cell boundary	Tuesday afternoon

K#	Brief	Filing date / phase
K281	Casey trio framework + inverted-pyramid leptons + emergent-vs-fundamental	Tuesday afternoon
K282	Grace parameter-reduction audit lens (program-level methodology)	Tuesday evening
K283 + v0.2	Lyra VEV no-form (walkback) + Elie correction VEV = 225·g substrate-primary RELABEL	Tuesday evening
K284	Elie mixing sector pre-check + 79-recurrence reduction candidate; Grace count-the-inputs bar	Tuesday evening
K285	Grace 4-category ledger framework + honest headline 2 of 26	Tuesday late-evening

7. Investigation Lanes Active (as of Tuesday midday June 9)

Lane	Primary CI	Status
Mixing sector K-type derivation (79-thread per F81)	Lyra	Multi-week opened Tuesday
F66 bulk-measure 225 forcing (VEV REDUCTION lead)	Lyra	Multi-week opened Tuesday via K283 v0.2
Filter 2 multi-week core (Yukawa<cell sharpened)	Lyra	Multi-week standing (K278 v0.2 + K280)
Tier 0 framework: action $\hbar$ _BST + two-time + spin magnitude	Lyra	Multi-week standing
Inverted-pyramid lepton K-type structure	Lyra	Opened K281 Tuesday
Parameter-reduction ledger (4-category operational)	Grace	Standing scorecard from K285
Yukawa hierarchy substrate-natural pattern map	Elie	Standing per K278 v0.2 + K284
Higgs investigation (Reading A vs Reading B)	Lyra + Grace + Elie	Multi-week per K281
Cascade a_{25} at n50	Elie	~2-3 days background
Vol 16 Ch 5 + Ch 8 multi-week prose absorption	Keeper	Standing multi-week
Cal cold-read backlog clearance	Cal	Substantial backlog

8. v0.2 absorption pending

v0.2 will absorb: - K286-K290 PRE-STAGE candidates as they fire (mixing sector outcomes, Tier 0 framework, lepton K-type, Higgs investigation) - Lyra F81 K-type derivation outcome - Grace ledger evolution - Methodology stack 20th-element ratification (K282 + K285) - Cal cold-read clearance notes - Cross-link map deepening as investigation lanes mature

Document utility: this registry serves as quick-navigation index for the K-audit chain at substantive substrate-architectural maturity. Use Section 6 (topical clusters) for thematic queries; Section 6 (full index) for K-number lookup; Section 4 (cross-link map) for substrate-architectural Schur generators + framework cross-references.

— Keeper, Tuesday 2026-06-09 midday (11:34 EDT date-verified; comprehensive K-audit registry filed per Casey directive; 115 K-audit files indexed across 6 phases + 10 topical clusters + 4 substantive cross-link threads; v0.2 absorption pending investigation evolution)